

Pathways of Thoughts: Multi-Directional Thinking for Long-form Personalized Question Answering

Alireza Salemi
University of
Massachusetts Amherst
Amherst, MA, USA
asalemi@cs.umass.edu

Cheng Li
Google DeepMind
Mountain View, CA, USA
chgli@google.com

Mingyang Zhang
Google DeepMind
Mountain View, CA, USA
mingyang@google.com

Qiaozhu Mei
University of Michigan
Ann Arbor, MI, USA
qmei@umich.edu

Zhuowan Li
Google DeepMind
Mountain View, CA, USA
zhuowan@google.com

Spurthi Amba
Hombaiah
Google DeepMind
Mountain View, CA, USA
spurthiah@google.com

Weize Kong
Google DeepMind
Mountain View, CA, USA
weize@google.com

Tao Chen
Google DeepMind
Mountain View, CA, USA
taochen@google.com

Hamed Zamani
University of
Massachusetts Amherst
Amherst, MA, USA
zamani@cs.umass.edu

Michael Bendersky
Google DeepMind
Mountain View, CA, USA
bemike@google.com

Abstract

Personalization is well studied in search and recommendation, but personalized question answering remains underexplored due to challenges in inferring preferences from long, noisy, implicit contexts and generating responses that are both accurate and aligned with user expectations. To address this, we propose *Pathways of Thoughts (PoT)*, an inference-stage method that applies to any large language model (LLM) without task-specific fine-tuning. PoT models the thinking as an iterative decision process, where the model dynamically selects among cognitive operations such as reasoning, revision, personalization, and clarification. This enables exploration of multiple reasoning trajectories, producing diverse candidate responses that capture different perspectives. PoT then aggregates and reweights these candidates according to inferred user preferences, yielding a final personalized response that benefits from the complementary strengths of diverse reasoning paths. Experiments on the LaMP-QA benchmark show that PoT consistently outperforms competitive baselines, achieving up to a 10.8% relative improvement. Human evaluation further validates these improvements, with annotators preferring PoT in 66% of cases compared to the best-performing baseline and reporting ties in 15% of cases.

CCS Concepts

• **Information systems** → **Question answering; Personalization; Computing methodologies** → **Markov decision processes; Natural language generation.**

Keywords

Personalization, Test-time compute scaling, Thinking LLMs

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1 Introduction

Personalizing large language models (LLMs) has become increasingly important due to applications in content generation, writing assistance, education, and recommendation [16, 18, 22, 23, 29]. By conditioning outputs on user preferences, context, and needs, personalization improves engagement, satisfaction, and overall effectiveness [28]. It is particularly valuable in information-seeking systems such as search engines and question answering (QA) [31, 32], where appropriate responses depend on user-specific factors like background knowledge and learning style. Despite its clear relevance to web search and emerging systems (e.g., AI-powered search modes), personalization in QA remains relatively underexplored.

Augmenting LLM inputs with the user’s personalized context have proven effective for personalized generation [16, 18, 27, 29]. This can be further enhanced by training the LLM to perform reasoning over the personalized context to extract the user’s preferences before response generation [28]. However, in most real-world cases, fine-tuning the LLM is often infeasible for two key reasons. On one hand, many LLMs are black boxes with frozen parameters, offering no control over their internal mechanisms. On the other hand, even with full access to parameters, the fine-tuning



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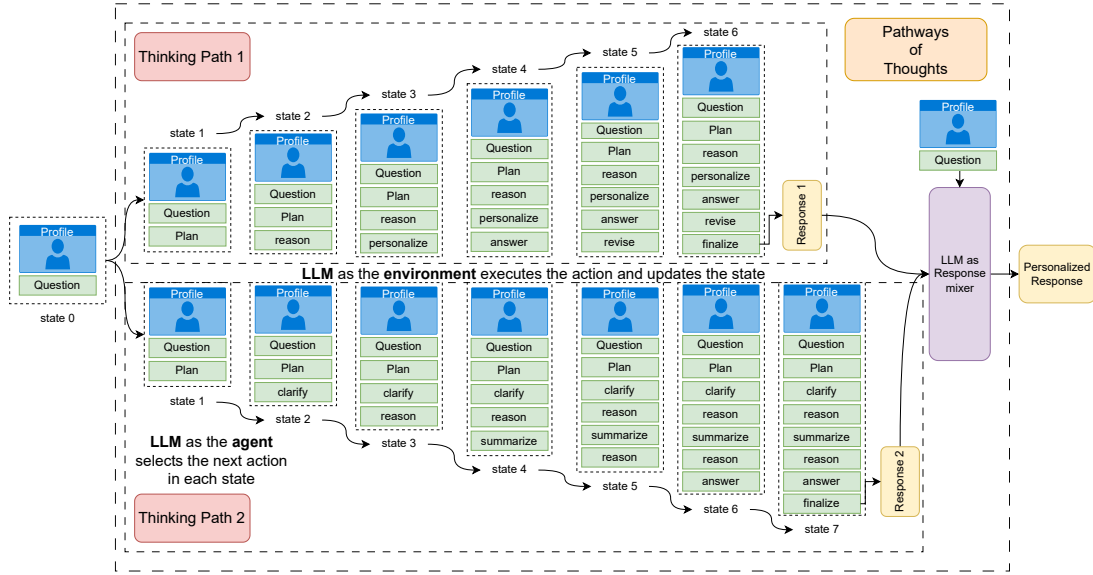


Figure 1: Pathways of Thoughts overview. PoT explores multiple directions of thinking about answering the question, generates a response for each, and mix them into a final response.

process requires substantial amount of personal data and computational resources for individual users. Consequently, a practical way to influence the model’s output is by applying advanced strategies during inference. Recognizing this, our paper focuses on this scenario and proposes an **inference-stage strategy** to achieve better personalization.

Developing effective personalization models presents its own unique challenges. First, the quality of a response in personalized QA is determined by the user who asks the question, which is influenced by their preferences, context, and background. Therefore, even when two answers are factually accurate, one may be preferred by the user. Second, user preferences are often not explicitly stated and must be inferred from personalized context, such as profiles or prior interactions. This context is frequently lengthy and noisy, making it difficult for a model to extract relevant signals. These issues highlight that simply augmenting an LLM’s input with user context is often insufficient. Effective personalization requires the model to perform reasoning over noisy, implicit user data to deduce user preferences before generating a response.

To address the aforementioned challenges of personalized QA, this paper introduces *Pathways of Thoughts (PoT)*, demonstrated in Figure 1. This approach, which is generic and can be applied to any LLM without additional training, enables the model to explore multiple thinking pathways when generating responses, each leading to a distinct response. We formalize each thinking pathway as a Markov Decision Process (MDP), where the LLM serves as both the agent and the environment. At each step, the LLM, acting as the agent, selects the most suitable action from a predefined set of cognitive operations, such as planning, reasoning, personalizing, and revising. The LLM then assumes the role of the environment, executing the chosen action and updating the state with newly acquired knowledge. This iterative process allows the model to refine its understanding of the user’s intent, extract key preferences from the user profile, and explore diverse strategies for answering the

question. The process continues until the model determines that no further actions are needed, producing an initial set of responses. To generate the final personalized response, the LLM mixes outputs from different thinking pathways into a single response, aligning them with the user’s preferences extracted from their profile. Pathways of Thoughts addresses the challenges of LLMs in personalized question answering by allowing the model to iteratively think about the next step in responding to the question and how the lengthy user context can be effectively utilized. Additionally, it explores different diverse directions and paths, considering a range of possible solutions for answering the question. Finally, generating a response by aggregating all pathways’ responses ensures that the output captures the strengths of all explored pathways, resulting in a more personalized and aligned response.

We conduct our experiments on the *LaMP-QA* benchmark [31]—a recent large-scale benchmark designed for evaluating personalized QA in three diverse domains: (1) Art & Entertainment, (2) Lifestyle & Personal Development, and (3) Society & Culture. Our results demonstrate that PoT outperforms competitive baselines across all categories, with up to 10.8% statistically significant improvement on average compared to the best-performing baselines. Additionally, we investigate the impact of pathway length, the number of pathways, and various methods for diversifying pathways and aggregating responses on performance. We show that PoT generalizes to different long-context backbone LLMs without requiring additional training. Finally, human evaluation of comparing outputs side by side shows that *PoT* better satisfies users’ information needs in 66% of cases, compared to 19% for the best-performing baseline.

2 Related Work

LLM Personalization. Personalization plays a central role in search, recommendation, and text generation applications [6, 23, 29, 39]. To enable personalization in LLMs, Salemi et al. [29] introduced

a retrieval-augmented generation framework alongside the LaMP benchmark for short-form text, which was later extended to long-form content generation with LongLaMP [16] and more recently to personalized QA with LaMP-QA [31]. Personalized assistants have also been studied in several recent works [18, 21, 22, 43]. Proposed approaches span a variety of techniques, including training retrieval models with user feedback [27], optimizing LLMs with personalized supervision [13], and generating personalized prompts [17]. Parameter-efficient fine-tuning has emerged as another direction [35], including its integration with retrieval-augmented methods [30]. Beyond model adaptation, reasoning and self-training strategies have been shown to improve personalization in text generation [28]. More recently, learning from natural language feedback has proven effective for adapting QA systems to individual users [32]. Nevertheless, strategies for improving personalized QA at test time—without incurring the cost of additional training—remain underexplored, which is the focus of this work.

Reasoners & Agents for Problem Solving. Reasoning enables models to solve problems with step-by-step processing, known as chain-of-thought (CoT), enhancing performance in complex tasks like mathematical problem-solving, logical reasoning, and common-sense understanding [19, 37, 42]. Training LLMs to perform intermediate reasoning steps before generating responses has been effective, as seen in GPT-o1 [25] and DeepSeek-R1 [5]. Additionally, agentic and multi-turn or multi-agent systems that interact with external tools [14], code [8, 20, 38], simulated environments [10, 26, 36, 41], or with other agents [9] have been explored for complex problem-solving. Markov Decision Processes (MDP) have shown promise in reasoning for games and mathematical problems [10, 36]. However, these methods have not been applied to free-form text generation, particularly personalization which requires user-specific responses rather than objectively correct answers. This study extends these methods to personalized text generation by integrating them with graph-based search in place of traditional tree search in MDPs, enabling the model to produce responses that reflect the aggregated quality of all explored candidate solutions rather than relying on a single path.

Scaling Inference Compute. Recent advances in LLM reasoning for logic and mathematics show that increasing compute resources during inference can significantly enhance performance [4, 33]. This allows LLMs to explore answer spaces more thoroughly, improving accuracy in logical reasoning, math problem-solving, and code generation [2, 3, 44]. While previous research has focused on math, coding, and logic, its potential for free-form text generation, especially personalized question answering, is underexplored. Our work addresses this by using increased inference compute to better navigate response spaces, enhancing personalization.

3 Problem Formulation

We assume a question x_u originates from a user u , who has a set of n_u personalized information elements, denoted as $P_u = \{p_i\}_{i=1}^{n_u}$. Following previous work that utilizes long-term user history as the user profile [15, 29, 31], P_u consists of the user’s previously asked questions along with the their question narrative. The objective is to use this personalized information, along with the question x_u , to

generate a personalized response using an LLM π , formulated as $\hat{y}_u = \pi(x_u, P_u)$. To evaluate the personalized response generated for the user u ’s question, following Salemi and Zamani [31], we assume access to a set of aspects $E_{x_u} = \{e_i\}_{i=1}^{|E_{x_u}|}$ that user u expects to be addressed in the response, extracted from a question narrative r_{x_u} written by the user. These aspects are used solely for evaluation and are not accessible to the model during response generation. Finally, a metric $\mu(x_u, \hat{y}_u, E_{x_u}, r_u)$ is used to score the generated response based on the extent to which these aspects are covered.

4 Pathways of Thoughts

Personalization is inherently user-specific, meaning that a response generation strategy effective for one user may not perform well for another. Consequently, relying on a single solution to generate a response for a user may not always yield the most preferred outcome for each individual user. To address this limitation, as well as those discussed in Section 1, we propose *Pathways of Thoughts (PoT)*, illustrated in Figure 1. In this framework, the LLM explores multiple thinking pathways, each corresponding to a distinct line of thought that leads to a potential response. We model each thinking pathway as an MDP, where the LLM serves as both the agent and the environment. At each step, the LLM (acting as the agent) selects an action from a set of fundamental cognitive operations such as planning, reasoning, clarification, or revising. The same LLM (now acting as the environment) then executes the chosen action, updating the current state with newly acquired information about the question and user. This iterative process continues until convergence, producing a final response for each pathway. To generate a single personalized response, the LLM aggregates the responses obtained from different pathways, integrating their complementary strengths while aligning with the user’s preferences inferred from their profile. The complete procedure is detailed in Algorithm 1, and the next section elaborates on this method.

4.1 Thinking as a Markov Decision Process

The thinking process in the human brain involves using and integrating a set of fundamental cognitive skills—e.g., reasoning, planning, and revision—that collectively form the foundation of structured thought [7, 34]. Inspired by theories about sequential decision making in the human brain, we model the thinking process of LLMs as an MDP. Given the current thoughts and steps taken to generate a response, the model selects the next fundamental action from a predefined set to enhance its representation of the question, refine its reasoning, and improve the generated response.

An MDP is defined as a quadruple (S, A_s, P_a, R_a) , where S is the set of states, A_s represents the set of available actions in state s_t , $P_a(s_{t+1} | s_t)$ defines the probability of transitioning from state s_t to s_{t+1} given action a_t , and $R_a(s_{t+1}, s_t)$ specifies the reward function, which assigns a reward for taking action a_t in state s_t and transitioning to s_{t+1} . In this setting, the agent π aims to assign a probability $p_\pi(a_t | s_t)$ to each action based on the current state s_t .

To generate a personalized response for the question x_u from the user u , we assume the agent π starts from an initial state s_0 , which consists of the question q_u and the user profile P_u . The agent iteratively selects the most probable action at each step using the prompt shown in Figure 7 in Appendix A, continuing this process

Initial State & Action Definition Prompt

Your task is to answer a user's question from a post in a personalized way by considering the user's past post questions and detailed descriptions of these questions. This is a multi turn task and you don't need to answer the question in one turn. In each turn, you can perform one action in that turn.

Your context information:

- The user's current question from a post.
- The user's past post questions and detailed descriptions of these questions.

Actions:

answer:

Definition: This action asks you to generate a personalized response to the question, if the question can benefit from personalization. Therefore, you should choose this action when you are ready to answer the question. This action should be executed when there is no ambiguity and all the information needed for answering the question is available.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- personalizedAnswer: a string representing your personalized answer to the user's current question considering the personalized context and previous taken steps.

check_personalization:

Definition: This action ask you to see if the question asked by the user can benefit from personalization or not. This action can be useful when you want to decide if you need to personalize the response or not.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- needs_personalization: a boolean that is true if the question can benefit from personalization and false if personalization cannot help.
- reason: a string representing the reason why the question can or cannot benefit from personalization.

personalize

Definition: This action asks you to find the relevant information about the question from the context and summarize how it can be helpful for answering the question. This action should be executed when you think personalized information is needed to answer the question.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- personalizedSummary: a string representing your summary of user context and how it is helpful to answer the question.

plan

Definition: This action asks you to generate a plan of steps that we can take to answer all the aspects of the question. This action should be executed when you think the the question has multiple aspects and planning future steps that you need to take to answer the question is necessary.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- plan: a string representing your plan of steps that we can take to answer the question.

reason

Definition: This action asks you to think step by step about one aspect of the question. This action should be executed when you know different aspects of the question and you want to think step by step about one aspect of the question and how it can be answered.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- aspect: a string representing the aspect of the question.
- reason: a string representing your reasoning of the question aspect and how it can be answered.

clarify

Definition: This action asks you to check if the question is ambiguous and how it can be clarified. This action should be executed when you think the current information needs more clarifying.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- ambiguous: a boolean indicating whether the question is ambiguous or not.
- clarification: a string representing your clarification questions of the current information and steps.
- clarification_response: a string representing your response to the clarification question. You should generate a reasonable response to clarification questions. This field should be empty if the question is not ambiguous.

summarize

Definition: This action summarizes all available information and findings about the question. This action should be executed when you think the current information needs more summarization.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- summarization: a string representing your summarization of the current information and steps.

revise

Definition: This action asks you to revise the current response to the question. This action should be executed when you provided an answer to the question and you want to revise the current response.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- revised: a string representing your revised response to the question.

finalize

Definition: This action will finalize the response to the question. This action should be executed when you think the response is finalized and no more action is required to answer or revise the answer to the question.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- personalizedAnswer: a string representing your finalized response to the question.

Your task:

This is a multi turn task and you don't need to answer the question in one turn. In each turn, you can perform multiple actions from an action list. Then, based on the detention of the action, you can do that action in that turn and provide the response. Therefore, your task in each turn consists of two steps:

Step 1: Selecting one of the actions that the user provides for you as the possible next actions. In this case, you should choose one of the actions considering that the main goal is generating a personalized response to the question. The action you choose should be the next best step toward the final goal, which is generating a personalized response to the question.

Step 2: In this step, based on the action definition, you can do that action in the same turn.

Output format:

You should generate a valid JSON, enclosed in ```json ``` block, with the following fields:

- action: a string representing your selected action. This text should be one of the valid actions that user provides.
- reason: a string representing the reason for why the action is selected and how it is helpful to reach the final goal, which is generating a personalized response to the question.
- actionOutput: a json representing the output of the action based on the defined output format for the selected action.

Your input:

Past post questions and detailed descriptions of these questions:

```
{profile}
```

Current post question:

```
{query}
```

Figure 2: Initial state and actions definition prompt used in Pathways of Thoughts.

until it either chooses a halting action (as determined by the LLM) or reaches the maximum number of allowed actions, denoted as T . Since our task involves text generation and lacks an external environment enforcing termination, the agent itself determines when to stop or continues until the predefined action limit is reached. See = Algorithm 1 (lines 3-17) for details. The following subsections outlines the definitions of each MDP component in our framework:

Set of States (S): The state is defined as the concatenation of the inputs, outputs, and decisions of the model into a sequence of tokens using a predefined prompt format. This representation captures the full context of the system like a conversation. The initial state of the system consists of the user's question x_u , the user profile P_u , and a set of instructions specifying the actions and the model's expected behavior for each action, as shown in Figure 2. Given this setup, the set of all possible states corresponds to the set of all textual sequences that this initial prompt is their prefix. As a result, the state space forms an uncountable discrete set.

Algorithm 1 Pathways of Thoughts Implementation using MDP and Global Search.**Input:** prompt x_u , profile P_u , max path length T , number of pathways N , backbone LLM π **Output:** personalized output $\hat{y}_{x_u}$

```

1:  $R_{\text{PoT}} = \{\}$   $\triangleright$  Set of all responses from different pathways
2: for  $i = 0$  to  $N$  do  $\triangleright$  Generate  $N$  pathways of thoughts
3:    $t = 0$ 
4:    $s_t = \text{init\_prompt}(x_u, P_u)$   $\triangleright$  Prompt in Figure 2
5:   while  $t < T$  do  $\triangleright$  Thinking for maximum  $T$  steps
6:      $A_{s_t} = \text{possible\_actions}(s_t, t)$   $\triangleright$  Find the possible
       actions for state  $s_t$ .
7:      $\bar{a}_p = \text{action\_selection\_prompt}(s_t, A_{s_t})$   $\triangleright$  Prompt in
       Figure 7
8:      $a_t = \pi(s_t; \bar{a}_p)$   $\triangleright$  Selecting the action (LLM as agent)
9:      $\hat{y}_t = \pi(s_t; \bar{a}_p; a_t)$   $\triangleright$  Performing the action (LLM as
       environment)
10:     $s_{t+1} = s_t; \bar{a}_p; a_t; \hat{y}_t$   $\triangleright$  new state by concatenating
       input/outputs in this time step
11:    if  $a_t = \text{finalize}$  then
12:       $R_{\text{PoT}} = R_{\text{PoT}} \cup \{\hat{y}_t\}$   $\triangleright$  Capturing the final response
       of this pathway
13:    break
14:    end if
15:     $t = t + 1$   $\triangleright$  updating the step variable
16:  end while
17: end for
18:  $\bar{I}_p = \text{important\_aspects\_prompt}(P_u)$   $\triangleright$  Prompt in Figure 8
19:  $I_u = \pi(\bar{I}_p)$   $\triangleright$  Extracting important aspects from user profile
20:  $\bar{y}_p = \text{mixture\_prompt}(I_u, x_u, R_{\text{PoT}})$   $\triangleright$  Prompt in Figure 10
21:  $\hat{y}_{x_u} = \pi(\bar{y}_p)$   $\triangleright$  Combining responses based on important
       aspects
22: return  $\hat{y}_{x_u}$   $\triangleright$  Returning final response

```

Set of Possible Actions for State s_t (A_{s_t}): We consider the following fundamental actions that the agent can choose from in thinking process: *answering*, *planning*, *personalization*, *personalizing*, *reasoning*, *clarifying*, *summarizing*, *revising*, and *finalizing*.¹ The detailed definition of each action and output format of them is provided in Appendix A.1. In the initial state, the agent is restricted to selecting the *planning* action, as it must first plan the steps required to address the user’s question. In subsequent states, the agent can choose from all the actions; however, the model can only select the *revising* action if it has previously selected the *answering* action, as revision typically follows the generation of a response.

Transitioning from State s_t to s_{t+1} with Action a_t ($P_{a_t}(s_{t+1} | s_t)$): We use the LLM as both the agent and the environment simultaneously: when it selects an action, it acts as the agent, and when it executes the action, it acts as the environment. This dual role allows the LLM to control the flow of thinking and performing it simultaneously. Accordingly, with the agent decision to perform action a_t at time step t , the LLM receives the state s_t and action a_t as input and generates the output $\hat{y}_t$. The concatenation of s_t , a_t ,

and $\hat{y}_t$ forms the next state s_{t+1} . Thus, the transition probability $P_{a_t}(s_{t+1} | s_t)$ is defined by the LLM itself.

Reward Function ($R_a(s_{t+1}, s_t)$): Since our agent is used without training, an explicit reward function is not required; we assume the agent makes reasonable choices. However, in general, an outcome reward function can be employed to assign rewards to actions based on the evaluation of the LLM’s final response to the question.

4.2 Multi-Directional Thinking by Diverse Planning and Mixture-of-Responses

We observed that when the agent selects *planning* as the first action, the subsequent steps consistently follow the outline of the generated plan. This suggests that encouraging multi-directional thinking about responding to the question can be achieved by generating a variety of plans, each exploring a different strategy for addressing the question. We study two methods for diverse planning:

- **Initial State Alteration:** If the initial state is modified before executing the planning action, the LLM can produce different plans, as the changes in personalized context influences the planning process. This can be modeled by varying the amount of past user interactions accessible to the agent, effectively addressing the challenge of long personal context. To implement this, we randomly select N subsets of the user profile P_u , each of size $\tau \times |P_u|$, where $0 \leq \tau \leq 1$. Starting from these N diverse initial states lead to different planning trajectories and thinking pathways.
- **Planning Action Variation:** Another way to generate different plans from the same initial state is to use a higher sampling temperature $0 \leq \tau \leq 1$ during plan generation, which increases the randomness in the generated plans, leading to more diverse outputs. Thus, we generate N different plans from the same initial state (created using the whole profile P_u) by setting a high temperature τ , leading to different thinking pathways.

Each thinking pathway results in a response, and together they form a set of response proposals $R_{\text{PoT}} = \{r_i\}_{i=1}^N$. To produce the final response $\hat{y}_{x_u}$ for the user, we need to ensure that it is the most suitable one for the user. To achieve this, we first use the LLM π to extract the important aspects I_u for user u from their user profile P_u using the prompt shown in Figure 8 in Appendix A.2. This serves as an intermediate step to learn about the user’s individual preferences. Based on this, there are two methods to generate the final response:

- **Best-of-N:** Given the extracted important aspects I_u for the user u and the input x_u , the LLM π selects the best response from the set of responses resulted from multi-directional thinking R_{PoT} , using the prompt shown in Figure 9 in Appendix A.2.
- **Mixture-of-N:** Given the extracted aspects I_u for the user u and the input x_u , the LLM π combines different aspects of the generated responses from the set of multi-directional thinking outputs R_{PoT} . The final response is denoted as $\hat{y}_{x_u} = \pi(x_u, R_{\text{PoT}}, I_u)$, using the prompt shown in Figure 10 in Appendix A.2. This mimics how humans solve problems by exploring multiple solutions and synthesizing a new solution that incorporates the strengths of them. This is beneficial for open-ended text generation, where

¹This action, *finalizing*, also referred to as halting, terminates the process.

Table 1: Performance on LaMP-QA. † and ‡ show statistically significant improvement over the single and multiple inference best baseline, respectively, using t-test ($p < 0.05$). Gemini 1.5 Pro has been used as the backbone LLM for all the methods.

Method	Arts & Entertainment	Lifestyle & Personal Development	Society & Culture	Average (macro)
Single Inference				
(1) No Personalization	0.1741	0.2863	0.2996	0.2533
(2) In-Context Personalization	0.1722	0.3022	0.3482	0.2742
(3) In-Context Personalization w/ CoT	0.2119	0.3862	0.4230	0.3403
(4) Pathways of Thoughts ($N = 1, T = 8$)	0.2499[†]	0.3966	0.4490[†]	0.3651[†]
Multiple Inference				
(5) In-Context Personalization w/ Best-of-32	0.2248	0.3753	0.4303	0.3434
(6) In-Context Personalization w/ CoT & Best-of-32	0.2607	0.4380	0.4789	0.3925
(7) In-Context Personalization w/ Mix-of-32	0.2310	0.3781	0.4411	0.3500
(8) In-Context Personalization w/ CoT & Mix-of-32	0.2689	0.4443	0.4891	0.4007
(9) Tree of Thoughts ($K = 32, B = 2$)	0.2054	0.4178	0.4546	0.3592
(10) Pathways of Thoughts ($N = 16, T = 8$)	0.2999^{†‡}	0.4960^{†‡}	0.5362^{†‡}	0.4440^{†‡}

there is no definitive right or wrong response. By aggregating the strengths of all responses, it generates a response that aligns with user preferences, ensuring a more personalized output. PoT uses this approach as the primary response aggregation method.

5 Experiments

5.1 Experimental Setup

Datasets. We conduct experiments on the LaMP-QA benchmark [31], the only publicly available dataset for long-form personalized QA. LaMP-QA covers three diverse domains: (1) Art & Entertainment (767 questions), (2) Lifestyle & Personal Development (989 questions), and (3) Society & Culture (1074 questions). Each LaMP-QA example contains a user question, the user’s question history (serving as the user profile), a user-written narrative that reflects the user’s information need and intent, and a set of personalized rubrics specifying the aspects an ideal response should address. For evaluation, we follow the evaluation recipe of Salemi and Zamani [31] with Gemini 1.5 Pro as the evaluator language model. For each question, the evaluator checks whether the generated response addresses each personalized aspect, assigning scores in the range $[0, 2]$. These are normalized to $[0, 1]$, and the final response score is computed as the mean normalized score across all aspects. For details, we refer the reader to Salemi and Zamani [31]. In one experiment, we corroborate our findings through a side-by-side human evaluation comparing PoT to the strongest baseline.

PoT’s Configurations. We use Gemini 1.5 Pro as the backbone LLM. To show the generalizability of PoT, we also use GPT-4o-mini² [24] using our most representative experiment configurations. For single pathways, we use Nucleus sampling [11] with a temperature of $\tau = 0.1$. For multi-directional pathways, we use the same setting, except for the *Planning Action Variation*, where a temperature of $\tau = 0.9$ is used exclusively for the *planning* action. For generating diverse plans, we use *Planning Action Variation* by default and *Mixture-of-N* for response aggregation, unless otherwise specified. We set the context length to $32k$, the maximum number of actions per pathway to $T = 8$, and the number of pathways to

$N = 16$. In ablations, we reduce the number of pathways to $N = 4$ to save costs, unless otherwise specified. For personalized context, we use the 10 recent questions of the user with the same order as they appear in the LaMP-QA [31] dataset.

Baselines. We use six baselines: (1) LLM without personalized context that directly answers the question, (2) LLM with access to the same personalized context as PoT, and (3) LLM with access to personalized context that first performs reasoning over the context to extract important aspects relevant to the question before generating the response. The two latter use the prompts shown in Figures 11 and 12 in Appendix B. These baselines all use Nucleus sampling with a temperature of $\tau = 0.1$. We extend the latter two baselines by doing multiple inferences and Best-of-N [3, 12] and Mixture-of-N to evaluate their performance when test-time compute scales, to form baselines (5), (6), (7), and (8). Additionally, we include the (9) Tree of Thoughts (ToT) [40] as a baseline, leveraging its test-time compute scaling for comparison with our method. We follow their setup, particularly in the creative writing task, as it is the closest task they studied to personalized question answering. This setup consists of an intermediate planning step, where a set of plans is generated and the best plan is selected, followed by a final response generation step, where a set of responses is generated and the best response is chosen. For planning, we use the prompts shown in Figure 13, and for response generation, we refer to Figure ??, both in Appendix B. We observed that the baselines generate, on average, half the number of tokens compared to PoT. To ensure a fair comparison, we set the Best-of-N and Mixture-of-N baselines to generate double the number of responses, with $N = 32$ and a sampling temperature of $\tau = 0.9$ to ensure diverse responses. For the Best-of-N and Mixture-of-N approach, we use the method introduced in Section 4.2. For ToT, following Yao et al. [40] and their setup for the creative writing task that is the most similar to ours, we set the tree depth to $B = 2$. For fair comparison with PoT, we generate $N = 32$ plans and responses at each node with a sampling temperature of $\tau = 0.9$. The prompts for selecting the best plan is shown in Figure ?? in Appendix B. For selecting the best response, we utilize the method introduced in Section 4.2.

²Available at: <https://platform.openai.com/docs/models/gpt-4o-mini>

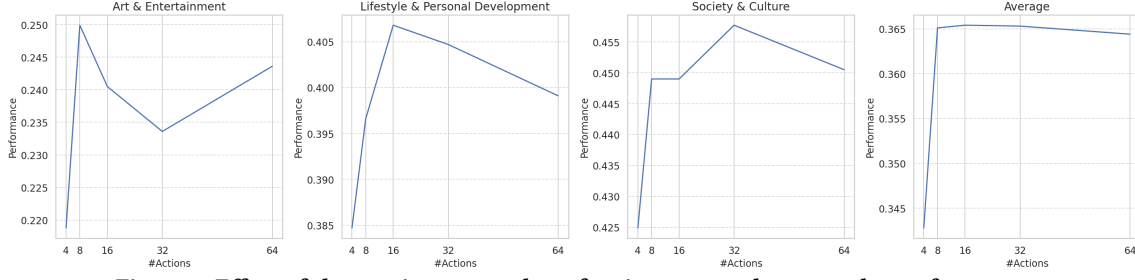


Figure 3: Effect of the maximum number of actions per pathway on the performance.

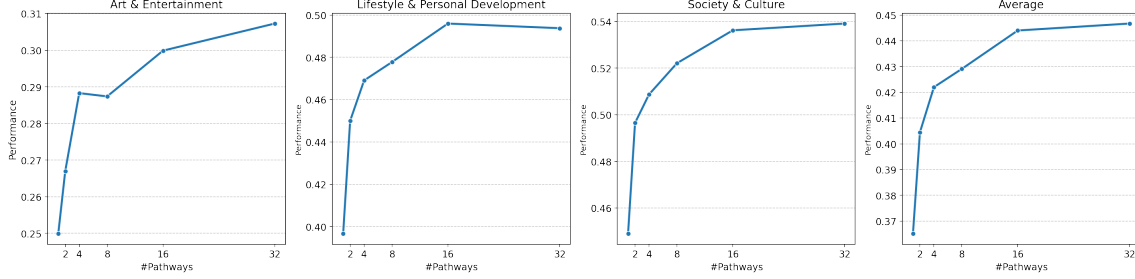


Figure 4: Effect of the number of pathways on the performance.

5.2 Empirical Findings

Comparison with baselines. The results of this experiment are reported in Table 1. The findings show that PoT outperforms all the baselines across all categories, achieving a **10.8%** relative improvement on average over the best baseline (row 8). The improvements are statistically significant with respect to t-test with 95% confidence. This demonstrates the superiority of PoT compared to the baselines. To compare our thinking-as-an-MDP method with baselines, we also report the results using only a single inference (rows 1-4). The results from this experiment show that our method (row 4) outperforms all single inference baselines (rows 1-3), with statistically significant improvements in two out of three categories and overall performance. Specifically, our method achieves a **7.2%** relative improvement in average performance, suggesting that our thinking approach is significantly more effective than CoT.

Effect of the maximum number of actions in PoT. For this, we focus on a single inference ($N = 1$) using PoT and vary the maximum number of actions it can perform. The performance across all categories is shown in Figure 3. The findings show that increasing the number of actions improves performance up to a certain point (8 actions per pathway), after which performance declines. This occurs because, with too few actions, the LLM cannot construct effective pathways to solve the problem. Conversely, when given too many actions, the LLM may overthink the problem, performing unnecessary actions that degrade performance [1].

Effect of the number of pathways in PoT. We set the maximum number of actions per pathway to $T = 8$ and vary the number of pathways. The performance across all categories is shown in Figure 4. The findings indicate that increasing the number of pathways consistently improves performance; however, as the number grows, the improvements become thinner. This occurs because there is a limited set of different ways for answering the question,

and after a certain point, the solutions start to repeat, leading to diminishing returns in performance with additional pathways.

Effect of different plan diversification and response aggregation in PoT. We evaluate all possible combinations of the methods introduced in Section 4.2. Specifically, we consider four configurations: 1) *Planning Action Variation w/ Best-of-N*, 2) *Planning Action Variation w/ Mixture-of-N*, 3) *Initial State Alteration w/ Best-of-N*, and 4) *Initial State Alteration w/ Mixture-of-N*. Additionally, we explore different values of the hyperparameter τ for both *Planning Action Variation* and *Initial State Alteration*. The results of this experiment are shown in Figure 5. Our analysis reveals several key observations. Most notably, *Mixture-of-N* consistently outperforms *Best-of-N* regardless of the plan diversification method. This can be attributed to the fact that *Best-of-N* selects only one response from the generated candidates, whereas *Mixture-of-N* enables the combination of strengths from multiple responses, resulting in a more comprehensive and higher-quality output. We observe that increasing the hyperparameter τ improves performance. In *Planning Action Variation*, a higher τ leads to a more diverse set of generated plans due to higher sampling temperature, resulting in more varied thinking pathways that may produce more tailored responses. In *Initial State Alteration*, increasing τ provides each pathway with more user-specific information, enabling more personalized responses. Comparing the two, *Planning Action Variation* generally outperforms *Initial State Alteration* in terms of average performance. However, as τ increases, the performance gap narrows. Overall, *Planning Action Variation w/ Mixture-of-N* achieves the best performance for personalized question answering.

PoT’s generalizability and independence of LLM. In the same setting as Gemini 1.5 pro, we apply PoT and the best performing baselines to another backbone LLM, GPT-4o-mini, to demonstrate its effectiveness across different LLMs with long context windows.

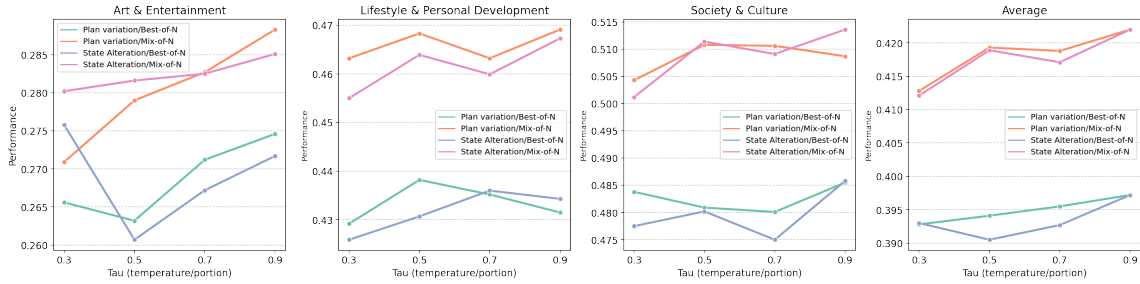


Figure 5: Effect of planning diversification and response aggregation methods on the performance.

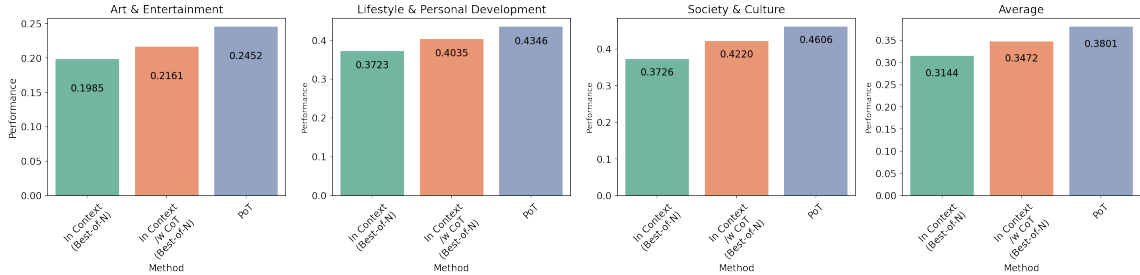


Figure 6: Performance of Pathways of Thoughts and baselines with GPT-4o-mini as the backbone LLM.

The average performance across all categories is presented in Figure 6. The results indicate that, similar to Gemini 1.5 Pro, GPT-4o-mini with PoT outperforms all baselines on all categories. It achieves a statistically significant 9.4% relative improvement on average across all datasets compared to the best baseline. These findings suggest that PoT enhances the performance of LLMs in personalized QA regardless of the underlying backbone LLM.

Human evaluation. We randomly sample 100 examples from LaMP-QA and generate outputs using PoT and In-Context Personalization w/ CoT & Best-of-32. Human annotators evaluate the outputs based on their alignment with the user’s question narrative (i.e., a detailed description of user’s information need and perspective). Each example is annotated twice, yielding an inter-annotator agreement Cohen’s kappa of 0.613. The results indicate that annotators preferred PoT in **66%** of the cases, the baseline in **19%**, and reported tie in **15%** of cases. These results show that PoT achieves considerably higher alignment with human preferences.

5.3 Case Study

We observe that PoT ($N = 16$, $T = 8$) generates 1943 unique pathways, defined by distinct action sequences chosen. Specifically, *Art & Entertainment* yields 1109, *Lifestyle & Personal Development* produces 814, and *Society & Culture* results in 1182 unique pathways. On average, each example generates 11.1 unique pathways with on average 7.19 actions. As a case study demonstrating the effectiveness of PoT, we present an example in Figure 16 in Appendix:

- **Thinking Pathways & Generated Responses:** To conserve space, we present in Figure 16 only the sequence of steps taken in each pathway, rather than the full generated text, and the response in each pathway. Each pathway corresponds to a distinct reasoning trajectory generated by PoT. For example, in the illustrated case, PoT selects different pathways depending on

whether personalization is required. Some pathways generate a direct response without personalization (e.g., Pathway 7), while others involve multiple rounds of reasoning that integrate various aspects of the query with the user profile before producing an answer (Pathways 1–6, 9–11, 13–15). Certain pathways also involve revision (Pathways 12, 15, and 16) or clarification (Pathway 4). Notably, we observed that for some trajectories PoT omit personalization entirely (Paths 7, 8, and 11). This shows that PoT deliberately explores a diverse range of reasoning strategies, producing responses from multiple perspectives that are subsequently combined into a final answer encompassing the advantages of each. Although responses across pathways share common elements, each retains distinctive characteristics, reflecting the variation in underlying reasoning strategies.

- **Final Response:** Figure 16 shows the final response to the user’s question, produced by aggregating and integrating components from multiple pathway outputs into a single coherent answer. The contributions of different pathways are indicated in brackets ([]). This shows how PoT leverages the complementary strengths of diverse reasonings, combining them into a unified response that aligns with the user’s preferences and key information needs.

6 Conclusions

This paper addresses the critical yet under-explored problem of personalized question answering. We propose *Pathways of Thoughts (PoT)*, a novel approach that models the LLM’s thinking process as an MDP during test time. By enabling the exploration of multiple thinking pathways and aggregating diverse responses, *PoT* effectively captures user preferences to generate personalized answers. Our results on *LaMP-QA* demonstrate that *PoT* achieves substantial improvements over baseline methods, with performance gains

of up to 10.8%. Furthermore, human evaluation confirms the effectiveness of our approach, showing a strong user preference over existing baselines. This paper focuses solely on personalized question answering; however Pathways of Thoughts is a general framework and future work can explore the potential of Pathways of Thoughts for long-form text generation beyond personalization.

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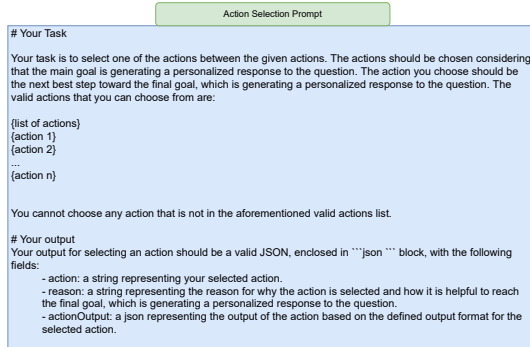


Figure 7: Action selection prompt used in PoT.

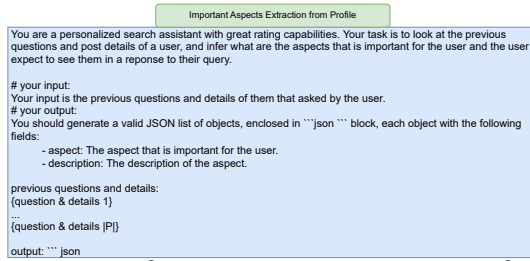


Figure 8: Prompt for extracting important aspects from the user profile used in PoT.

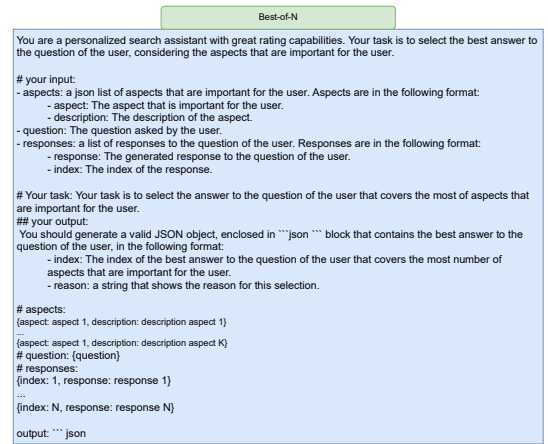


Figure 9: Prompt for selecting the best response out of N generated responses used in PoT.

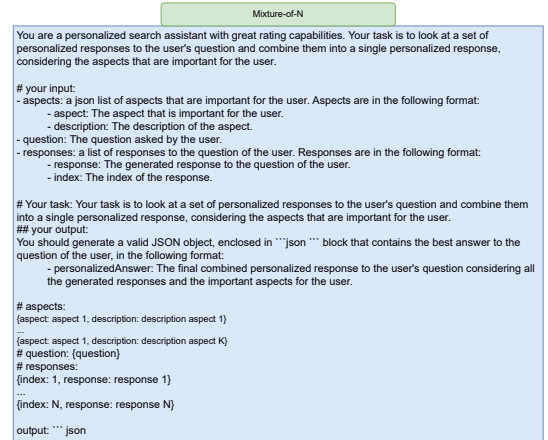


Figure 10: Prompt for mixing the N generated responses into a single response used in PoT.

A PoT's Implementation Details

A.1 Actions Definition

We define an action as a means by which the agent interacts with the text generation environment, specifically the LLM, by instructing it to perform a particular operation. Each action is associated with a specific output format that the LLM generates in response to the instruction. In this paper, we consider the following actions as the operations the model can take to respond to a question:

- **answering:** This action instructs the model to generate a personalized response to the question, provided that personalization can enhance the response. It should be selected when the model is prepared to answer the question, ensuring that all necessary information is available and there is no ambiguity.
- **planning:** This action instructs the model to generate a plan of steps required to address all aspects of the question. It should be executed when the question has multiple components, and the model recognizes that planning the next steps is necessary to adequately respond to the user's query.
- **personalization:** This action asks the model to evaluate whether the user's question can benefit from personalization. It is useful

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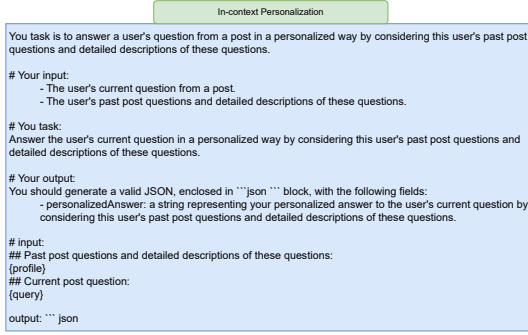


Figure 11: Prompt for the In-Context Personalization.

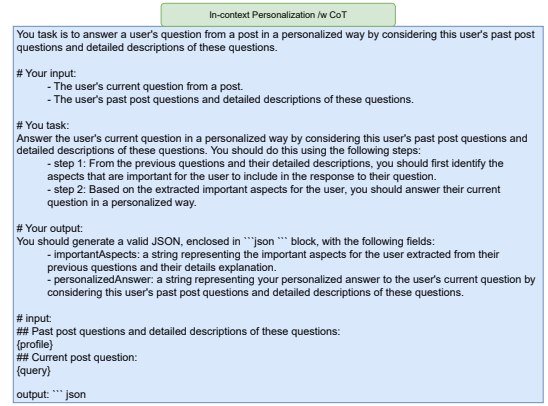


Figure 12: Prompt for In-Context Personalization with CoT.

when determining whether personalizing the response is necessary or if a generic response would suffice.

- *personalizing*: This action asks the model to identify relevant information from the user profile and summarize how it can be used to answer the question. It is executed when personalized information is necessary to enhance the quality of the response.
- *reasoning*: This action prompts the model to break down one aspect of the question and think through it step by step. It is executed when questions involve multiple aspects, and the model needs to focus on and consider how to address a specific aspect.
- *clarifying*: This action prompts the model to assess whether the question is ambiguous and determine how it can be clarified. It is executed when the model identifies that the current information is insufficient or unclear, requiring further elaboration.
- *summarizing*: This action instructs the model to summarize all available information and findings. It should be executed when the model determines that the current information needs to be consolidated or synthesized to ensure a clearer understanding before continuing with the response generation.
- *revising*: This action instructs the model to revise the current response to the question. It should be executed when the model has already generated an initial response but believes the response can be improved or adjusted to better address the user's query or to incorporate additional insights or information.
- *finalizing*: This action finalizes the response. It should be executed when the model believes the response is complete, accurate, and addresses the user's query. No further actions or revisions are necessary, and the response is ready to be presented.

A.2 Details of Response Aggregation Methods

An essential component of Pathways of Thoughts involves aggregating the candidate responses generated across different thinking pathways into a single output. Each pathway produces a candidate response, collectively forming a set of response proposals, denoted as R_{PoT} . To generate the final response $\hat{y}_{x_u}$ personalized for the user, it is crucial to select the most suitable candidate. To achieve this, we first employ the LLM π to extract the key aspects I_u relevant to user u from their user profile P_u . This extraction process is guided by the prompt illustrated in Figure 8, acting as an intermediate step to capture the user's individual preferences. Based on this, there are two methods to generate the final response:

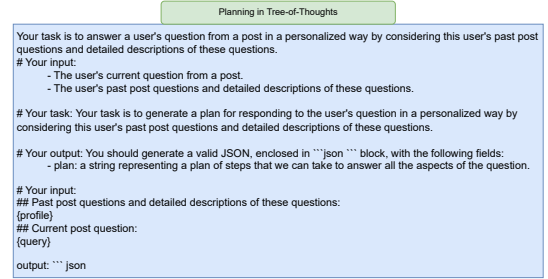


Figure 13: Prompt for the planning step in ToT.

- *Best-of-N*: Given the extracted important aspects for the user and the input, the LLM selects the optimal response from the set of candidate responses generated through multi-directional thinking. This is guided by the prompt shown in Figure 9.
- *Mixture-of-N*: Given the extracted important aspects for the user and the input, the LLM integrates different aspects of the candidate responses from the set of multi-directional thinking outputs, guided by the prompt in Figure 10. This method emulates human problem-solving by exploring multiple solutions and synthesizing a new response that leverages the strengths of all candidates.

B Baselines Prompts

Figures 11 and 12 present the prompts used for the personalized baselines without and with chain-of-thought reasoning, respectively. For the Tree of Thoughts baseline, Figure 13 shows the planning prompt, Figure ?? illustrates the response generation prompt, and Figure ?? provides the plan selection prompt.

Combined Response with Mixture-of-16

It's commendable that you're being mindful of the dynamics within your workplace, especially given that it's minority-led. Bringing up political concerns in any workplace can be tricky, but your past experiences with grant writing, advocating for accessibility, and navigating complex situations while traveling have equipped you with valuable skills for this conversation. Here's a combined approach, drawing from the best advice and considering your emphasis on clear, concise, and respectful communication: **[[Preparation and Understanding:]]** **[[Clarify your goals:]]** What do you hope to achieve by bringing up this topic? Are you seeking change, raising awareness, or simply expressing your concerns? **[[Response 5]]** **[[Research:]]** Understand the business's stance on the issue and consider how your views align with their values. **[[Response 5]]** **[[Contextualize:]]** Is the topic directly relevant to the business's operations, values, or industry? How might it affect employees, customers, or the business's reputation? **[[Response 1]]** **[[Company Culture:]]** Is your workplace generally open to discussing political matters? **[[Response 10]]** **[[Framing the conversation:]]** **[[Focus on shared values:]]** Frame your concerns in a way that aligns with the company's mission and values, emphasizing common ground and potential benefits for the business. **[[Responses 2, 4, 5, 6, 14]]** **[[Use respectful language:]]** Avoid inflammatory rhetoric, personal attacks, and emotional outbursts. Stick to the facts and present your concerns calmly and respectfully. **[[Responses 1, 7, 11, 12]]** **[[Be a good listener:]]** Engage in active listening and be open to different perspectives, even if you disagree. **[[Responses 1, 2, 4, 5, 6, 9, 11, 13]]** **[[Offer solutions:]]** Instead of just raising concerns, actionable steps the business could take. **[[Responses 5, 10, 14]]** **[[Choosing the Right Approach:]]** **[[Open dialogue with management:]]** If you feel comfortable, consider having a private conversation with your manager or a trusted supervisor. **[[Response 1]]** **[[Create a safe space for discussion:]]** Suggest a facilitated discussion group or an open forum where employees feel comfortable sharing their thoughts. **[[Response 1]]** **[[Written communication:]]** Consider a well-structured letter or email outlining your concerns and suggestions if you prefer a more formal approach. **[[Response 1]]** **[[Choose the right time and setting:]]** Avoid bringing up politics during busy work hours or in inappropriate settings. Opt for a private meeting or a time when open dialogue is encouraged. **[[Responses 2, 6, 9, 10, 11, 13, 14]]** **[[Remember:]]** **[[Power dynamics:]]** Be mindful of your position within the company and the potential impact of your words, especially in a minority-led business. **[[Responses 3, 6, 8, 11, 12]]** **[[Be prepared for different outcomes:]]** The business leadership might be receptive or dismissive. Consider how you will respond in each scenario, maintaining your composure and advocating for your needs respectfully. **[[Response 2]]** **[[Know when to disengage:]]** If the conversation becomes unproductive or heated, it's okay to politely disengage and revisit the topic at another time. **[[Response 1, 9]]** **[[Company policies:]]** Familiarize yourself with any workplace policies regarding political discussions. **[[Responses 3, 11, 8]]** By approaching the conversation with sensitivity, respect, and a willingness to listen, you can create a more productive and positive dialogue, even on potentially sensitive topics. Your experience with clear and concise communication will be a valuable asset in navigating this conversation effectively.

Figure 16: A case study of Pathways of Thoughts with $N = 16$ and $T = 8$ for question: What's the best way for me to bring up a topic of political concern if the business I work for is minority-led?

